

Reinforcement Learning Laboratory Experiment 1

A PROJECT REPORT

Submitted by

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LAB TASK - 1

In this task, the Gymnasium library was used to create and explore the CartPole-v1 Reinforcement Learning environment. The environment was initialized, observation and action spaces were analysed, and a random agent was executed for one complete episode. The agent selected random actions, and the observations, rewards, and episode termination status were displayed to understand the basic interaction between an agent and its environment.

CODE IMPLEMENTATION

```
main.py > ...
1  import gymnasium as gym
2  import numpy as np
3  import matplotlib
4
5  # Create environment
6  env = gym.make("CartPole-v1")
7
8  # Reset environment
9  observation, info = env.reset()
10
11 print("Environment Created Successfully!\n")
12
13 print("Initial Observation:")
14 print(observation)
15
16 print("\nObservation Space:")
17 print(env.observation_space)
18
19 print("\nAction Space:")
20 print(env.action_space)
21
22 print("\nObservation Space Type:")
23 print(type(env.observation_space))
24
25 print("\nNumber of Possible Actions:")
26 print(env.action_space.n)
27
28 print("\n=====")
29 print("Running Random Agent")
30 print("=====")
31
32 step = 0
33 total_reward = 0
```

main.py > ...

```
34
35     terminated = False
36     truncated = False
37
38     while not (terminated or truncated):
39
40         # Select random action
41         action = env.action_space.sample()
42
43         # Execute action
44         observation, reward, terminated, truncated, info = env.step(action)
45
46         step += 1
47         total_reward += reward
48
49         print(f"\nStep {step}")
50         print("Action      :", action)
51         print("Observation :", observation)
52         print("Reward       :", reward)
53         print("Terminated  :", terminated)
54         print("Truncated   :", truncated)
55
56     print("\n=====")
57     print("Episode Finished")
58     print("=====")
59
60     print("Total Steps :", step)
61     print("Total Reward:", total_reward)
62
63     env.close()
```

OUTPUT:

```
(venv) PS E:\MTech\Sem3\RL\RL-Lab-1> python main.py

Initial Observation:
[ 0.04372855  0.00459881 -0.00550211  0.00431114]

Observation Space:
Box([-4.8          -inf -0.41887903      -inf], [4.8          inf 0.41887903      inf], (4,), float32)

Action Space:
Discrete(2)

Observation Space Type:
<class 'gymnasium.spaces.box.Box'>

Number of Possible Actions:
2
```

The CartPole-v1 environment was successfully created and initialized using the Gymnasium library. After resetting the environment, the initial observation was displayed as four numerical values representing the cart position, cart velocity, pole angle, and pole angular velocity. The observation space is of type **Box**, indicating that the observations are continuous values. The action space is **Discrete(2)**, which means the agent can perform only two actions: move the cart to the left (0) or to the right (1). The observation space type confirms that the environment uses continuous state values, while the number of possible actions is two.

```
=====
Running Random Agent
=====

Step 1
Action      : 1
Observation : [ 0.04382052  0.19979924 -0.00541589 -0.2901027 ]
Reward      : 1.0
Terminated  : False
Truncated   : False

Step 2
Action      : 0
Observation : [ 0.04781651  0.00475493 -0.01121794  0.00086723]
Reward      : 1.0
Terminated  : False
Truncated   : False

Step 3
Action      : 0
Observation : [ 0.04791161 -0.19020435 -0.0112006   0.28998974]
Reward      : 1.0
Terminated  : False
Truncated   : False
```

A random agent interacts with the CartPole environment by selecting a random action at every time step. For each step, the environment returns the updated observation, the reward received, and the episode status. The reward is **1.0** for every successful step in which the pole remains balanced. The **Terminated** value is **False**, indicating that the pole has not fallen, and **Truncated** is **False**, indicating that the maximum episode length has not been reached. This process continues until the episode ends.

```
Step 19
Action      : 0
Observation : [-0.07672331 -0.59764826  0.20570765  1.2786335 ]
Reward      : 1.0
Terminated  : False
Truncated   : False

Step 20
Action      : 1
Observation : [-0.08867628 -0.40565428  0.23128033  1.0567666 ]
Reward      : 1.0
Terminated  : True
Truncated   : False

=====
Episode Finished
=====
Total Steps : 20
Total Reward: 20.0
```

The episode ended after **20 steps** because the pole could no longer be balanced, resulting in **Terminated = True**. Since the episode ended due to failure and not because of reaching the maximum step limit, **Truncated** remained **False**. The random agent achieved a **total reward of 20.0**, which indicates that it successfully balanced the pole for 20 consecutive time steps. The total reward is equal to the number of successful steps because the environment provides a reward of **1.0** for each step in which the pole remains balanced.

Conclusion

The task successfully demonstrated the basic interaction between an agent and the CartPole-v1 environment using the Gymnasium library. The agent observed the environment, selected random actions, received rewards, and continued interacting until the episode terminated. This experiment helped in understanding the fundamental concepts of Reinforcement Learning, including **environment initialization, observation space, action space, rewards, episode termination, and the agent-environment interaction loop**.